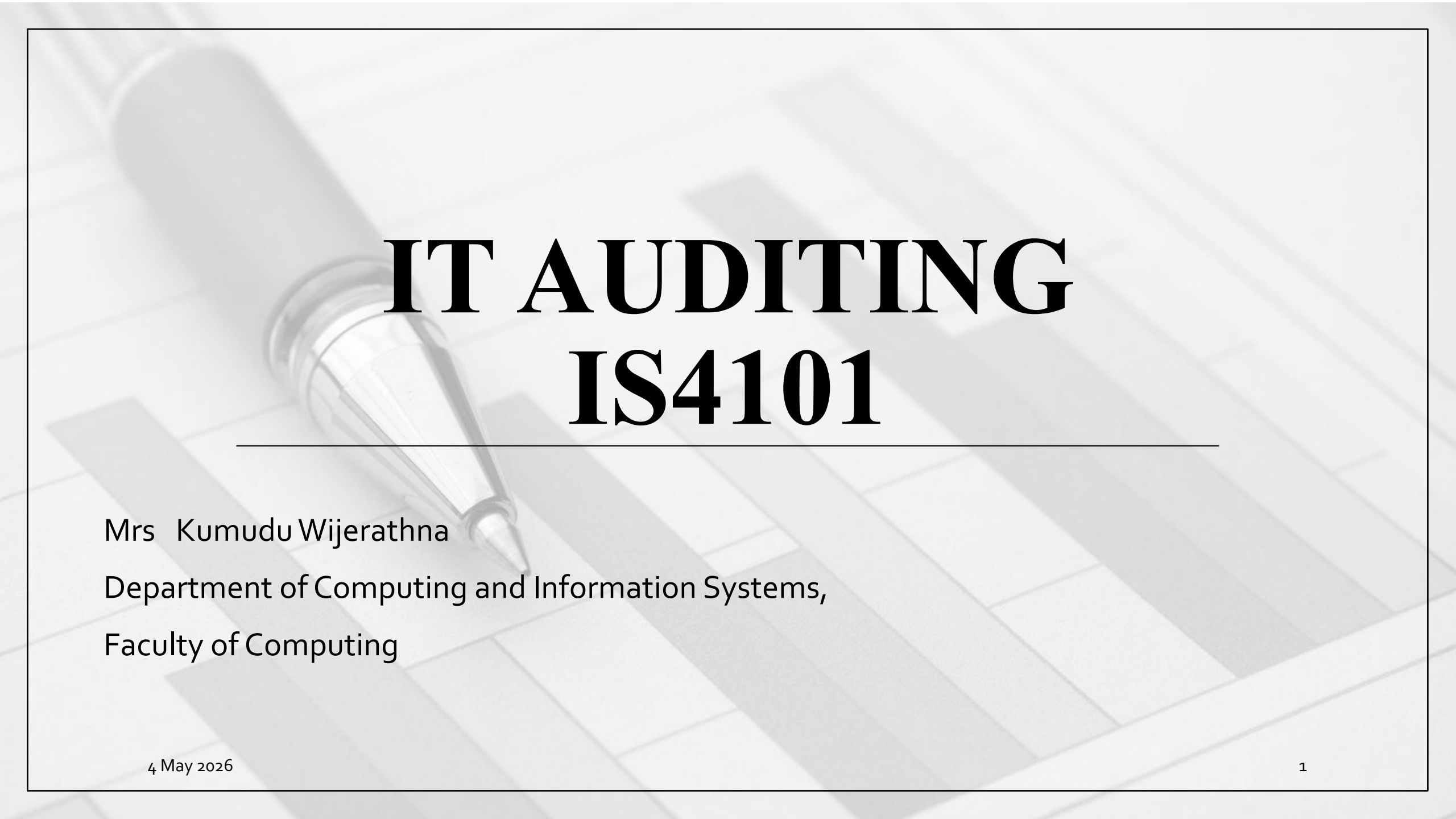


The background of the slide features a grayscale image of a silver ballpoint pen resting diagonally across a spreadsheet with a grid of cells.

IT AUDITING IS4101

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Lesson Learning Outcome

By the end of this lesson, you will be able to,

- Apply basic audit procedures for OS controls

Content

Auditing Operating Systems and Networks

Operating System in IT Auditing

- The operating system: the computer's control program

Operating system objectives

1. Translation of high-level languages:

Converts high-level programming languages into machine-level code for execution.

2. Resource allocation:

Distributes computer resources among users, workgroups, and applications.

3. Task management:

Handles job scheduling and multiprogramming for efficient operation.

Five Fundamental Control Objectives of the OS

- 1. Protection from Users:** Prevents user applications from gaining control or damaging the OS, ensuring continued operation and data integrity.
- 2. Protection of Users from Each Other:** Ensures that one user cannot access, destroy, or corrupt another user's data or programs.
- 3. Protection of Users from Themselves:** Prevents different modules within a user's application from interfering with each other, protecting the integrity of the data and programs.

Five Fundamental Control Objectives of the OS

4. Self-Protection of the Operating System: Safeguards individual OS modules from being destroyed or corrupted by other modules.

5. Protection from the Environment: In case of power failures or disasters, the OS should manage a controlled shutdown, allowing for recovery after the issue is resolved.

Operating System Security

- Operating system security includes the policies, procedures, and controls that define who can access the OS, which resources (e.g., programs, printers) they can use, and what actions they are permitted to take.

Operating System Security

Key Security Components in Secure Operating Systems

- **Log On Procedure:** The process users follow to authenticate themselves and gain access to the system.
- **Access Token:** A security credential used to identify the user and define their privileges.
- **Access Control List (ACL):** A list that specifies which users or system processes can access particular resources and what operations they can perform.
- **Discretionary Access Privileges:** Allows users to control access to the resources they own or manage.

Operating System Controls & Audit Tests

- **Importance of OS Integrity**

- If the integrity of the operating system is compromised, controls within individual accounting applications, especially those affecting financial reporting, may also be compromised.
- Auditors focus on the following areas to preserve operating system integrity:
 - a) Access privileges,
 - b) Password control
 - c) Virus control
 - d) Audit trail control

a. Controlling Access Privileges

- Access privileges are assigned to individuals and workgroups authorized to use the system.
- Scope : privileges determine access to directories, files, applications, and resources, as well as the types of actions that can be taken.
- Impact on security : the way access privileges are assigned influences overall system security.
- Privileges should be carefully managed and monitored for compliance with organizational policies and internal control principles.

Audit objectives: verify that access privileges are consistent with the separation of incompatible functions and adhere to organizational policy.

Audit Procedures:

1. Policy Review:

Assess the organization's policies for separating incompatible functions to ensure they support reasonable security.

2. Privilege Review:

Examine user groups' and individuals' privileges to ensure they align with job descriptions and positions. Confirm access is granted on a need-to-know basis.

3. Personnel Review:

Check if privileged employees undergo thorough security clearance checks as per company policy.

4. Log On Times Review:

Ensure users' log on permissions match the tasks being performed.

b. Password Control

A password is a secret code used by users to access systems, applications, data files, or network servers.

- **Function:**
 - The operating system denies access if the correct password is not provided.
- **Common Security Issues:**
 - Forgetting Passwords: Leads to being locked out of the system.
 - Infrequent Password Changes: Reduces security.
 - "Post-it Syndrome": Writing down passwords, making them visible to others.
 - Simplistic Passwords: Easily guessed by attackers.

Password Control Options

- Reusable Passwords:
 - The most common method of password control, where the same password is used repeatedly.
- One-Time Passwords:
 - Designed to overcome issues with reusable passwords by providing a new password for each session.

Audit Objectives:

- Ensure the organization has an adequate and effective password policy for controlling access to the operating system.

Audit Procedures:

1. Password Requirement Verification:

Confirm that all users are required to have passwords.

2. User Instruction:

Verify that new users are instructed on the use and importance of passwords.

3. Password Change Procedures:

Review procedures to ensure passwords are regularly changed.

4. Password Standards Assessment:

Assess the adequacy of password standards, including length and expiration intervals.

5. Weak Password Identification:

Review the password file to identify and disallow weak passwords, possibly using scanning software.

6. Password File Encryption:

Verify that the password file is encrypted and that the encryption key is properly secured.

c. Controlling Against Malicious and Destructive Programs

Malicious programs cause significant corporate losses annually.

Leading to:

- Data Corruption & Destruction
- Degraded Computer Performance
- Hardware Destruction
- Privacy Violations
- Significant Personnel Time for Damage Repair

Audit Objective

- Prevention Focus:
 - The key to controlling computer viruses and malicious programs is prevention through strict adherence to organizational policies and procedures.
- Audit Goal:
 - Verify that effective management policies and procedures are in place to prevent the introduction and spread of destructive programs, including viruses, worms, back doors, logic bombs, and Trojan horses.

Audit Procedures

1. Personnel Awareness:

Conduct interviews to confirm that operations personnel are educated about computer viruses and understand the risks of practices that can introduce or spread malicious programs.

2. Software Testing:

Verify that new software is tested on standalone workstations before being implemented on the host or network server.

3. Antiviral Software:

Confirm that the current version of antiviral software is installed on the server and that upgrades are regularly downloaded to workstations.

d. System Audit Trail Controls

- System audit trails are logs that record the activities of users and processes within the operating system, providing a historical record of actions and events.
- Audit trails help in monitoring, detecting unauthorized access, and identifying issues within the system.

Audit Objective

- Ensure that audit trails are comprehensive, accurately maintained, and effectively used for monitoring system activity and detecting security breaches.

Audit Procedures

1. Review Audit Trail Configurations:

Verify that audit trails are properly configured to capture all relevant activities, including user logins, access to sensitive data, and administrative actions.

2. Examine Log Integrity:

Assess the integrity of the audit logs to ensure they are tamper-proof and securely stored.

3. Monitor Audit Trail Usage:

Verify that the audit trails are regularly reviewed by authorized personnel to detect and respond to suspicious activities.

4. Retention Policy:

Check that there is a clear retention policy for audit trails, ensuring logs are kept for an appropriate period and archived securely.

Summary

- Operating system is critical for system control and security
- Strong OS controls protect data and users
- Passwords, access control, and antivirus are key security measures
- Audit trails help monitor system activities
- Auditors ensure OS controls are effective and secure

Thank you